# REVIEW REPORT

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| **Manuscript Title** | Sucralose and Liver Cancer: A Molecular Insight into the Computational Assessment Lurking in Everyday Sweeteners |
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| **Article Type** | Original Research Article |
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## Overview of Manuscript

The manuscript entitled **"Sucralose and Liver Cancer: A Molecular Insight into the computational assessment Lurking in Everyday Sweeteners"** investigates the potential association between sucralose, a widely used artificial sweetener, and liver cancer through a comprehensive network toxicology and molecular modeling approach. The study identifies common molecular targets between sucralose and hepatocellular carcinoma (HCC), constructs protein–protein interaction networks, and performs GO/KEGG enrichment analyses to reveal relevant biological pathways. Molecular docking and 100 ns molecular dynamics simulations further demonstrate stable interactions of sucralose with key cancer-related proteins such as CASP3, MAPK1, and PTGS2. The findings suggest possible mechanisms through which sucralose may influence carcinogenic pathways, while emphasizing the need for experimental validation to confirm these computational predictions.

## Highlights of the Study

- **Study Focus:** Investigated the potential molecular link between sucralose consumption and liver cancer development.
- **Target Identification:** Identified 60 overlapping targets between sucralose-associated proteins and liver cancer-related genes.
- **Network Analysis:** Constructed a protein–protein interaction network and identified six major hub genes (CASP3, GAPDH, MAPK1, MMP9, PTGS2, VEGFA).
- **Pathway Enrichment:** GO and KEGG analyses revealed involvement in apoptosis, inflammation, immune regulation, and cancer-related pathways.
- **Microarray Validation:** GEO datasets confirmed MAPK1 and PTGS2 as significantly upregulated genes in liver cancer.
- **Survival Significance:** Kaplan–Meier analysis demonstrated clinical relevance of key hub genes in hepatocellular carcinoma.
- **Molecular Docking:** Sucralose exhibited strong binding affinities toward PTGS2 (−7.2 kcal/mol), MAPK1 (−6.7 kcal/mol), and CASP3 (−5.7 kcal/mol)
- **MD Simulation:** 100 ns molecular dynamics simulations confirmed stable protein–sucralose complexes with favorable RMSD, RMSF, and hydrogen-bond profiles

## Highlights of the Study

- **Mechanistic Insights:** Proposed that sucralose may influence apoptosis, inflammation, angiogenesis, and tumor progression pathways.
- **Public Health Relevance:** Provides computational evidence supporting further toxicological and epidemiological investigations into artificial sweetener safety.
- **Integrated Computational Framework:** Combined network toxicology, differential gene expression analysis, molecular docking, survival analysis, and molecular dynamics simulations to comprehensively evaluate the potential hepatocarcinogenic effects of sucralose.
- **Future Research Foundation:** Established a scientific basis for future experimental, toxicological, and epidemiological studies aimed at assessing the long-term safety and cancer-related risks of artificial sweeteners in humans.

## Significance and Originality

The study is particularly significant because it bridges the gap between food safety assessment and cancer biology by applying advanced computational toxicology tools to a commonly consumed artificial sweetener. Through the integration of network toxicology, transcriptomic validation, molecular docking, and molecular dynamics simulations, the research provides a systems-level understanding of how sucralose may interact with critical oncogenic and inflammatory pathways in hepatocellular carcinoma. The identification of MAPK1, PTGS2, and CASP3 as potential molecular mediators offers novel hypotheses for future laboratory and clinical investigations. These findings not only contribute to the growing field of predictive toxicology but also provide valuable evidence that may support future regulatory evaluations, risk assessment strategies, and public health policies regarding long-term artificial sweetener consumption.

### Comments for Editor

The manuscript addresses an important and timely public health concern regarding the long-term safety of artificial sweeteners. The integration of network toxicology, molecular docking, gene expression validation, and molecular dynamics simulations strengthens the scientific value of the study. The methodology is comprehensive and the results are presented systematically. However, the conclusions are based entirely on computational analyses, and the manuscript would benefit from additional experimental validation to confirm the biological relevance of the predicted interactions. Minor language editing is also recommended to improve clarity and readability. Subject to these considerations, the study contributes meaningful insights into the potential molecular mechanisms linking sucralose exposure and hepatocellular carcinoma and is suitable for publication.

**Final Recommendation:** ✅ Accepted for Publication — No Revisions Required